



ECONOMIC RESEARCH
FEDERAL RESERVE BANK OF ST. LOUIS
WORKING PAPER SERIES

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Working Paper Number	2024-027F
Revision Date	October 2025
Citable Link	https://doi.org/10.20955/wp.2024.027
Suggested Citation	Bick, A., Blandin, A., Deming, D.J., 2025; The Rapid Adoption of Generative AI, Federal Reserve Bank of St. Louis Working Paper 2024-027. URL https://doi.org/10.20955/wp.2024.027

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The Rapid Adoption of Generative AI*

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October 27, 2025

Abstract

Generative artificial intelligence (AI) is a potentially important new technology, but its impact on the economy depends on the speed and intensity of adoption. This paper reports results from a series of nationally representative U.S. surveys of generative AI use at work and at home. As of late 2024, 45% of the U.S. population age 18-64 uses generative AI. Among employed respondents, 27% used generative AI for work at least once in the previous week: 10% used it every workday, and 17% on some but not all workdays. Relative to each technology's first mass-market product launch, work adoption of generative AI has been as fast as the personal computer (PC), and overall adoption has been faster than either PCs or the internet. Between 1 and 7% of all work hours are currently assisted by generative AI, and respondents report time savings equivalent to 1.4% of total work hours. Potential productivity gains vary widely by industry, and firm climate and policies play an important role in adoption patterns.

JEL Codes: J24, O33

Keywords: Generative AI, Technology Adoption, Labor Productivity

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1 Introduction

Generative artificial intelligence (genAI) has quickly emerged as a potentially important workplace technology. The large language model ChatGPT debuted in November 2022 and, by December 2024, had over 300 million weekly users. Several studies find that genAI boosts productivity in specific cases (Brynjolfsson et al., 2023; Cui et al., 2024; Dell’Acqua et al., 2023; Demirci et al., 2025; Dillon et al., 2025; Kanazawa et al., 2025; Noy and Zhang, 2023; Otis et al., 2024; Peng et al., 2023), and some project genAI will have large macroeconomic effects (Chui et al., 2023; Filippucci et al., 2024). Yet others predict only modest aggregate impacts (Acemoglu, 2024). These differing views largely reflect uncertainty about the extent of genAI adoption today and in the future.¹

This paper studies the adoption of genAI using data from the first nationally representative U.S. surveys of genAI usage at work and at home. Our data come from the Real-Time Population Survey (RPS), a national online labor market survey of working age adults aged 18-64 that has run since 2020 (Bick and Blandin, 2023). We report the combined results of two surveys fielded in August and November 2024, which collectively included more than 10,000 respondents. These surveys include questions about genAI adoption that we adapt from the Current Population Survey’s Computer and Internet Supplement (CIU), which was introduced in 1984 to document computer (and later, internet) adoption. We also introduce novel questions to measure how intensively individuals use the technology, which tasks genAI assists with, how much time it saves them, and workplace barriers or support for adoption. The resulting microdata will be made publicly available, addressing calls for more research that develops and shares novel data (Ethiraj et al., 2019).

We use this data to address three main questions. First, how many individuals are currently using genAI? Pooling our August and November 2024 surveys, 32% of workers reported using genAI for work: 10% used it every work day, 17% on some but not all work days, and 5% did not use it in the previous week. One third of respondents report using genAI outside of work. Overall, 45% of respondents report using genAI either for work or outside of work. The most commonly used products are ChatGPT (32% of all respondents), Gemini (19%) and embedded products such as Microsoft Copilot (15%).

To benchmark these results, we show that the rate of workplace genAI adoption in 2024, two years after the release of ChatGPT, is similar to the rate of workplace computer adoption in

¹Acemoglu et al. (2022) analyze online job vacancies related to AI, but their data predates ChatGPT. Albanesi et al. (2024), Eisfeldt et al. (2024), and Gathmann et al. (2024) compare occupations according to predictions of AI exposure but do not have data on actual adoption. Humlum and Vestergaard (2025) find no meaningful impacts of genAI adoption on employment and earnings in Denmark among workers in 11 occupations that are particularly exposed to genAI.

1984, three years after the release of the IBM PC (the first mass-market computer). The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption thus far suggests that it has the potential to be similarly disruptive. Moreover, we show that genAI adoption by workers is proceeding at a faster pace than adoption by firms, highlighting both challenges and opportunities for managers seeking to capitalize on this new technology (Babina et al., 2024; Bonney et al., 2024; Doshi and Moore, 2025; McElheran et al., 2024).²

Second, which workers are currently using genAI, and what factors contribute to an individual’s adoption decision? We find substantial variation in genAI adoption across demographic and labor market characteristics. Adoption has been faster for male, younger, more educated, and higher-wage workers. Adoption also differs greatly by occupation, suggesting that differences in job roles and responsibilities are an important determinant of adoption. Consistent with this, we show that recent estimates of genAI task-based exposure are highly predictive of actual adoption (Eloundou et al., 2024; Felten et al., 2021), and that occupations with lower adoption also have higher shares of workers who report that genAI is not useful for their job. These findings contribute to our understanding of how genAI maps onto specific tasks and roles within the firm, which is relevant both for allocating adoption support and adapting firm organization.

We provide evidence that firm climate and policies also appear to play an important role in workers’ adoption choice. A sizable share of workers who do not use genAI cite workplace barriers as a key reason, and this share is particularly large in a few occupations who may deal with sensitive data. Alternatively, workers whose employers encourage them to use genAI are much more likely to do so, even after controlling for other characteristics. This suggests that when job tasks are amenable to genAI adoption, firms can play an active role by promoting usage, offering training, and removing organizational barriers to adoption.

Third, how large are potential productivity gains from genAI? Based on self-reported time savings due to genAI, the average genAI user would need to work 5.2% more hours in order to complete their work without genAI, implying 1.4% time savings across all workers (including non-users). We use a standard production model to estimate potential productivity gains at the industry and aggregate level. We find wide variation in potential productivity gains by industry,

²The Census BTOS Survey estimates between August and November 2024 5.4% of firms had incorporated AI technologies into their production process. This suggests that firm-based measures may miss an important share of genAI adoption. Aaron Levie, the founder and CEO of Box, a cloud storage provider, compares this pattern with past technologies: “It’s almost a one-to-one parallel with the early wave of consumerization of IT. We saw that exact same dynamic in the early 2010s, which was employees of enterprises were pretty frustrated with their traditional document management and collaboration technology, and so they would go out and just go onto a web browser, sign up for a Box account, and then bring it into their organization.” (Levie, 2024)